

Subarnaduti Paul

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Summary

Ph.D. candidate specializing in continual learning algorithms, with expertise in computer vision, multimodal AI, and Bayesian uncertainty modeling. Driving research on the sequential adaptation of foundational models through advances in continual learning and machine unlearning algorithms, with a strong focus on scalability and computational efficiency. Passionate about advancing interpretability and fairness in large-scale models, and eager to contribute to both academic and applied real-world applications.

Education

Technical University of Munich, MSc in Communication Engineering Oct 2018 – Nov 2021

- **DGPA:** 1.7/1.0
- **Coursework:** Deep Learning, Applied Machine Intelligence, Computer Vision, Multi-View Geometry
- **Thesis:** "Autoencoder-based Efficient Feature Extractor for Noisy Industrial Data" (**Grade: 1.0**)

Bengal Institute of Technology, Kolkata, India, B.Tech in Electronics & Communication Engineering Aug 2014 – Jun 2018

- **DGPA:** 1.6 (8.91/10)
- **Coursework:** Engineering Mathematics I & II, Data Algorithms, C Programming, Java, DSP, Wireless Communication

Experience

Ph.D. Researcher, TU Darmstadt & University of Bremen, Germany Jul 2022 –

- Researching scalable continual learning algorithms for foundational models with reduced computational complexity.
- Proposed novel continual learning algorithms for Federated systems (**CONFEDMADE**), and multimodal setups (**CoreTokensets**), coupled with Variational Continual Learning.
- Developed a benchmark for machine unlearning in LLMs, revealing key weaknesses of language models in handling semantically evolving linguistic concepts.

Visiting Student Researcher, RIKEN AIP, Tokyo, Japan June & Nov – Dec 2025

- Investigating scalable approaches for Variational Continual Learning in foundational models, focusing on uncertainty quantification with Bayesian algorithms.

AI Engineer Intern, Bosch, Ulm, Germany Feb 2021 – Feb 2022

- Designed an unsupervised feature extraction algorithm for industrial sensory data, tackling key challenges such as noisy input data, costly labeling, and complex input environment.
- Researched and proposed a unique attention mechanism for 1D time-series data.
- Successfully implemented a pipeline to train and deploy models online on edge devices using cloud services such as Microsoft Azure.

Software Engineering Intern, BMW, Munich, Germany Jun 2019 – Oct 2020

- Integration, testing, and flashing of RSE builds on BMW vehicles.

Research Intern, Technical University of Munich May 2019 – Aug 2019

- Designed and simulated switching features for on-chip network routers (ONoC) within the Chair of Electronic Design Automation.

Publications

- 2025- "CHRONOBERG: Capturing Language Evolution and Temporal Awareness in Foundation Models". Preprint (**Joint first author**)
- 2024 - "CoreTokensets for Data Efficient Sequential Adaptation of Transformers". Published at 6th CLVision Workshop, ICCV, 2025. (**first author**)
- 2023 – “Masked Autoencoders are Efficient Continual Federated Learners.” Published at COLLAS 2024. (**first author**)
- 2025- The Cake that is Intelligence and Who Gets to Bake it: An AI Analogy and its Implications for Participation. Preprint. (**co-author**)
- 2024 - "BOWL: A Deceptively Simple Open World Learner". Published at CoLLAS 2025 (**co-author**)
- 2022 – “Extract, Compress, and Encode: LitNet, an Autoencoder for Noisy Time-Series Data.” Published at ICIT22, China. (**Joint first author**)

Projects

- Continually merging LLMs based on Fisher Information Matrix (FIM)** May-June 2025
- Investigated ways to sequentially merge models in an online continual setup by computing its relevancy via FIM.
- From Samples to Tokens: Building compact memory buffers** Apr-June 2024
- Designed a token-level data summarization algorithm for transformer architectures by interpreting the attention scores across the input tokens.
- Sustaining Reduction in Carbon Emissions Based on Policy Decisions post COVID-19, TUM** Apr 2020
- Built a web app implementing data analysis and ML pipelines with interactive visualization.
- Radio-Controlled Multi-Functional Clock in Zynq-7000 FPGA, TUM** Dec 2019
- Designed RTL architecture for Global FSM and integrated with alarm/stopwatch on FPGA.
- Hardware-Oriented Implementation of IDEA in Spartan-3e FPGA, TUM** Aug 2019
- Implemented the International Data Encryption Algorithm in VHDL and optimized scheduling.

Technologies

- **Languages:** Python, C, C++ , MATLAB, VHDL, Verilog, Assembly, JavaScript, HTML, CSS.
- **Frameworks:** PyTorch, TensorFlow, Keras, OpenCV
- **Tools & Version Control:** Docker, Git, LaTeX, CI/CD Pipeline, Xilinx ISE, Multisim, Microsoft Office.
- **Database & Cloud:** SQL, Microsoft Azure
- **Operating Systems:** Linux, Windows